

Variational Analysis/Numerical Optimization PhD Exam, January 2018

Do 8 (eight) problems.

1. Let $\mathbf{Q}$ denote an n by n symmetric matrix and let $\mathbf{d}_k$, $1 \leq k \leq n$, denote a collection of directions that are $\mathbf{Q}$ -conjugate.

- (a) Suppose the $\mathbf{Q}$ is positive definite. Show that the directions $\mathbf{d}_k$, $1 \leq k \leq n$, are linearly independent, and express the solution of the linear system $\mathbf{Q}\mathbf{x} = \mathbf{b}$ in terms of the conjugate directions.
- (b) If $\mathbf{d}_k^T \mathbf{Q} \mathbf{d}_k > 0$ for each k , then show that $\mathbf{Q}$ is positive definite.

2. Consider the following variational problem:

$$\min \left\{ \int_0^1 \frac{1}{2} u'(x)^2 + e^{u(x)} dx : u \in H^1([0, 1]), \quad u(0) = u(1) = 0 \right\}.$$

- (a) What is the first-order necessary optimality condition (Euler equation) for this variational problem?
- (b) Consider a uniform mesh $x_k = kh$ where $h = 1/N$; let u_k denote an approximation to $u(x_k)$. Of course, $u_0 = u_N = 0$. Give a finite difference approximation in terms $u_1, \dots, u_{N-1}$ to the solution of the Euler equation.
- (c) Let $\mathbf{F}(\mathbf{u})$ denote the finite difference system where $\mathbf{u} = (u_1, u_2, \dots, u_{N-1}) \in \mathbb{R}^{N-1}$, and let $\mathbf{u}^*$ denote the vector formed by evaluating the solution of the Euler equation at the mesh points $x_1, x_2, \dots, x_{N-1}$. Obtain a bound for the components of $\mathbf{F}(\mathbf{u}^*)$ in terms of the mesh spacing h .

3. Consider the following variational problem where f is a given smooth function:

$$\min \left\{ \Phi(u) := \int_0^1 \frac{1}{2} u'(x)^2 + f(x)u(x) dx : u \in H^1([0, 1]), \quad u(0) = u(1) = 0 \right\}.$$

Let S^h denote the piecewise linear finite element space defined on a uniform mesh with $v^h(0) = v^h(1) = 0$ for all $v^h \in S^h$. Obtain a bound for the error in the finite element approximation u^h obtained by minimizing Φ over S^h .

4. Suppose that $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is twice continuously differentiable, $\mathbf{x}^* \in \mathbb{R}^n$, $\nabla f(\mathbf{x}^*) = 0$, and $\nabla^2 f(\mathbf{x}^*)$ is positive definite. Show that $\mathbf{x}^*$ is a strict local minimizer of f .
5. Suppose that a quadratic $\Phi(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} - \mathbf{b}^T \mathbf{x}$ is minimized by steepest descent: $\mathbf{x}_{k+1} = \mathbf{x}_k - s_k \mathbf{g}_k$, where $\mathbf{g}_k = \nabla \Phi(\mathbf{x}_k)$ and s_k is the stepsize.
 - (a) Derive the formula for the Cauchy step.
 - (b) Derive the formula for the BB step.
6. Consider the system of differential equations $\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t), \mathbf{u}(t))$, $\mathbf{x}(0) = \mathbf{x}_0$, where $\mathbf{f} : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n$ and $\mathbf{f}$ satisfies the global Lipschitz condition

$$\|\mathbf{f}(\mathbf{x}_1, \mathbf{u}_1) - \mathbf{f}(\mathbf{x}_2, \mathbf{u}_2)\| \leq L(\|\mathbf{x}_1 - \mathbf{x}_2\| + \|\mathbf{u}_1 - \mathbf{u}_2\|).$$

Show that the differential equation has a global solution and on any interval $[0, T]$, the following bound holds for all $(\mathbf{u}_1, \mathbf{u}_2) \in \mathcal{L}^\infty([0, T])$:

$$\|\mathbf{x}_1 - \mathbf{x}_2\|_{\mathcal{L}^\infty} \leq e^{LT} \|\mathbf{u}_1 - \mathbf{u}_2\|_{\mathcal{L}^1},$$

where $\mathbf{x}_i$ is the solution of the differential equation associated with $\mathbf{u}_i$, $i = 1, 2$.

7. Suppose that P and Q are continuous on $[0, 1]$ and the following functional is nonnegative over all $h \in H_0^1([0, 1])$:

$$\Omega(h) = \int_0^1 (P(x)h'(x)^2 + Q(x)h(x)^2) dx.$$

Show that $P \geq 0$ on $[0, 1]$.

8. Suppose that $u : [0, 1] \rightarrow \mathbb{R}$ is twice continuously differentiable, and let u^I be the continuous, piecewise linear interpolant of u . Prove the following bound:

$$\|u - u^I\|_{\mathcal{L}^\infty} \leq \frac{h^2}{8} \|u''\|_{\mathcal{L}^\infty}.$$

9. Suppose that $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a convex function.

(a) If f is continuously differentiable, then show that for all $\mathbf{x}$ and $\mathbf{y} \in \mathbb{R}^n$, we have

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})(\mathbf{y} - \mathbf{x}).$$

(b) If f is twice continuously differentiable, then show that the Hessian $\nabla^2 f(\mathbf{x})$ is positive semidefinite for all $\mathbf{x} \in \mathbb{R}^n$.

10. Suppose that $\Phi : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and $\mathbf{x}^* \in \mathbb{R}^n$ is a fixed point of Φ .

(a) If Φ is a contraction mapping with contraction constant λ , then show that the iteration $\mathbf{x}_{k+1} = \Phi(\mathbf{x}_k)$ converges to $\mathbf{x}^*$ from any starting guess $\mathbf{x}_0$, and

$$\|\mathbf{x}_k - \mathbf{x}^*\| \leq \lambda^k \|\mathbf{x}_0 - \mathbf{x}^*\|.$$

(b) Suppose that Φ is continuously differentiable on a convex set $\mathcal{K} \subset \mathbb{R}^n$. Let μ denote the supremum of the singular values of $\nabla \Phi$ over $\mathcal{K}$. Show that

$$\|\Phi(\mathbf{x}) - \Phi(\mathbf{y})\| \leq \mu \|\mathbf{x} - \mathbf{y}\|$$

for all $\mathbf{x}$ and $\mathbf{y} \in \mathcal{K}$.